

Day 4: Polymer Chain MD Simulation with GROMACS

This tutorial demonstrates how to prepare a solvated polymer system and perform molecular dynamics (MD) simulations using **GROMACS**. The workflow includes energy minimization, NVT equilibration, and NPT equilibration.

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Initialization

The first step is to set up the environment for the tutorial. This includes:

- Opening the terminal and navigating to the `polymermd-workshop-main` folder:

```
cd /PATH_TO/polymermd-workshop-main
```

Replace `PATH_TO` with the directory path on your computer.

- Activating the conda environment:

```
conda activate polymer-md
```

- Launching Jupyter Lab:

```
jupyter lab --ip 0.0.0.0 --no-browser
```

Open the printed URL in your browser, or configure it to open automatically.

- Installing required Python packages:

```
pip install ngview parmed pandas panedr MDAnalysis
```

System Generation

Copy Initial Files

Copy the polymer structure (**.gro**) and topology (**.top**) files generated in Day 3 to the current working directory.

```
polymer_name = "PE025_PET25_random"
polymer_dir = os.path.join(PROJECT_ROOT, "simulations", polymer_name)
os.makedirs(polymer_dir, exist_ok=True)
os.chdir(polymer_dir)

# Copy necessary files
copy_gro = f"cp ../../Day3-gen-polymer-
solved/polymers/{polymer_name}/polymer.gro polymer.gro"
copy_top = f"cp ../../Day3-gen-polymer-
solved/polymers/{polymer_name}/topol.top topol.top"
run_command(copy_gro)
run_command(copy_top)
```

Solvate the Polymer

Add a cubic water box around the polymer and solvate the system with SPC/E water molecules.

```
editconf_cmd = "gmh editconf -f polymer.gro -o polymer_box.gro -d 1.0 -bt
cubic"
run_command(editconf_cmd)

solvate_cmd = "gmh solvate -cp polymer_box.gro -cs spc216.gro -o
polymer_solv.gro -p topol.top -maxsol 1000"
run_command(solvate_cmd)
```

Add Ions

Neutralize the system and add Na⁺ and Cl⁻ ions to achieve a concentration of 0.01 M.

```
grompp_cmd = "gmh grompp -f minimization.mdp -c polymer_solv.gro -p
topol.top -o ions.tpr"
run_command(grompp_cmd)

genion_cmd = "echo SOL | gmh genion -s ions.tpr -o polymer_solv_ions.gro -
p topol.top -pname NA -nname CL -neutral -conc 0.01"
run_command(genion_cmd)
```

Energy Minimization

Perform energy minimization to remove steric clashes and optimize the system geometry.

```
em_dir = os.path.join(PROJECT_ROOT, "simulations", polymer_name, "em")
os.makedirs(em_dir, exist_ok=True)
os.chdir(em_dir)

# Write energy minimization parameters
with open("em.mdp", "w") as f:
    f.write("""
integrator = steep
emtol      = 10.0
emstep     = 0.01
nsteps     = 100000
nstlist    = 1
ns_type    = grid
rlist      = 1.3
rcoulomb   = 1.3
rvdw       = 1.3
pbc        = xyz
""")

# Generate .tpr file and run minimization
grompp_cmd = "gmh grompp -f em.mdp -c ../polymer_solv_ions.gro -p
../topol.top -o em.tpr -maxwarn 2"
run_command(grompp_cmd)

mdrun_cmd = "gmh mdrun -s em.tpr -deffnm em -v"
print(f"Run the following command in the terminal to start energy
minimization:\n{mdrun_cmd}")
run_command(mdrun_cmd)
```

Equilibration

NVT Equilibration

Stabilize the system at constant number of particles (N), volume (V), and temperature (T).

```
nvt_dir = os.path.join(PROJECT_ROOT, "simulations", polymer_name, "nvt")
os.makedirs(nvt_dir, exist_ok=True)
os.chdir(nvt_dir)

# Write NVT equilibration parameters
with open("nvt.mdp", "w") as f:
    f.write("""
```

```

integrator          = md
nsteps             = 50000
dt                 = 0.002
tcoupl             = V-rescale
tc-grps            = System
tau_t              = 0.1
ref_t              = 300
pcoupl             = no
gen_vel            = yes
gen_temp           = 300
""")

# Generate .tpr file and run NVT equilibration
grompp_cmd = "gmh grompp -f nvt.mdp -c ../em/em.gro -p ../topol.top -o
nvt.tpr -maxwarn 2"
run_command(grompp_cmd)

mdrun_cmd = "gmh mdrun -s nvt.tpr -deffnm nvt -v"
print(f"Run the following command in the terminal to start NVT
equilibration:\n{mdrun_cmd}")
run_command(mdrun_cmd)

```

NPT Equilibration

Stabilize the system at constant pressure (1 bar) and temperature (300 K).

```

npt_dir = os.path.join(PROJECT_ROOT, "simulations", polymer_name, "npt")
os.makedirs(npt_dir, exist_ok=True)
os.chdir(npt_dir)

# Write NPT equilibration parameters
with open("npt.mdp", "w") as f:
    f.write("""
integrator          = md
nsteps             = 50000
dt                 = 0.002
tcoupl             = V-rescale
tc-grps            = System
tau_t              = 0.1
ref_t              = 300
pcoupl             = Berendsen
pcoupltype         = isotropic
tau_p              = 2.0
ref_p              = 1.0
compressibility     = 4.5e-5
""")

# Generate .tpr file and run NPT equilibration
grompp_cmd = "gmh grompp -f npt.mdp -c ../nvt/nvt.gro -p ../topol.top -o
npt.tpr -maxwarn 2"
run_command(grompp_cmd)

```

```
mdrun_cmd = "gmx mdrun -s npt.tpr -deffnm npt -v"  
print(f"Run the following command in the terminal to start NPT  
equilibration:\n{mdrun_cmd}")  
run_command(mdrun_cmd)
```

Visualization

Visualize the polymer structure or trajectory using **nglview**.

```
visualize_trajectory("polymer_solv_ions.gro")
```

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